INSTRUCTIONS

MASTER SWITCHES

IC3012BH600 AND IC3012BH700

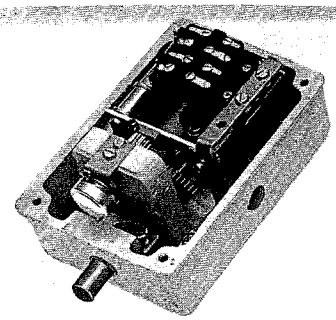


Fig. 1. IC3012BH600-699 Foot-operated

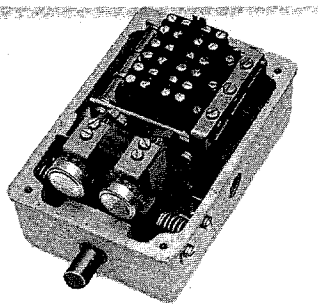


Fig. 2. IC3012BH600-699 Hand-operated

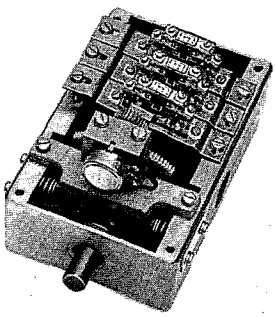


Fig. 3. IC3012BH-700-729 Hand-or foot-operated

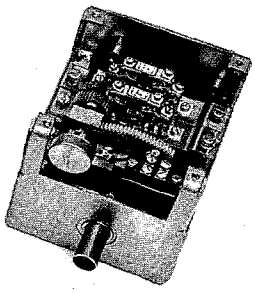


Fig. 4. IC3012BH730-759 Foot-operated

Before any adjustments, servicing, parts replacement or any other act is performed requiring physical contact with the electrical working components or wiring of this equipment, the BATTERY AND DISCHARGE CAPACITOR 1C MUST BE DISCONNECTED.

DESCRIPTION

The IC3012BH600 and the IC3012BH700 represent a family of accelerator master switches that are either foot-operated through a pedal and linkage system, or hand-operated by a suitable handle arrangement. They function to provide speed control for a battery truck traction motor by varying the resistance in an SCR control system. As the switch shaft is rotated, a segment gear mounted on the shaft drives a pinion gear, which in turn rotates a potentiometer. With the potentiometer at high ohms, the truck operates at "creep" speed. As the potentiometer is rotated towards zero ohms, the truck approaches top SCR speed smoothly.

This family of accelerators also contains either two or three switchette units that are operated by cams mounted on the main switch shaft. These cams are designed to a variety of contours to provide a predetermined control sequencing arrangement for control functions.

Enclosures are provided for general heavy industrial application.

CURRENT RATING OF SWITCHETTES

Voltage Inductive	Current-Amperes	
	Make and Break	Carry
6	10.0	10
12	6.0	10
18	4.0	10
24	3.5	10
30	3.0	10
36	2.5	10
48	2.0	10
60	1.2	10
72	1.0	10

The two basic types of switchettes used are the single-pole, double-throw (1 N.O. – 1 N.C.) and the double-pole, double-throw (2 N.O. – 2 N.C.).

INSTALLATION

This family of accelerators is designed for mounting as indicated in the following table.

Type IC3012-	Four-point Bottom Mounting	Three-point Bottom Mounting
BH600's	X	
BH700-729		X
BH730		X

The information contained herein is intended to assist truck users and dealers in the servicing of control furnished by the General Electric Company. It does not purport to cover all details or variations in equipment or provide for every possible contingency to be met in connection with installation, operation or maintenance.

Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the truck manufacturer through his normal service channels, not directly to General Electric Company.

The IC3012BH600 series is designed to facilitate a conduit connection or wire harness leaving the enclosure.

The IC3012BH700 series is designed for either of the above methods of connection, plus a plug-in receptacle located in the side wall of the enclosure.

Spring-return or maintained-position forms are also available. The spring-return forms are designed for hand operation. When switches are used for foot operation it is necessary to provide an external spring to overcome the friction build-up of the connecting linkage. It is also recommended to locate an adjustable mechanical stop under the foot pedal. The stop will terminate the travel of the pedal and linkage system rather than transmitting the force to the internal stops of the accelerator switch.

MAINTENANCE

Oil-less bearings are used on both ends of the main operating shaft, thus they require no attention. The switches using metallic pinion and segment gears should be greased periodically with some Gredag* type lubricant. The switch using nonmetallic parts to drive the potentiometer require no lubrication. No other parts require lubricating. If the potentiometer becomes inoperative, it can be removed by unsoldering the attached wires, loosening the clamp which secures the pinion gear to the potentiometer shaft, and removing the potentiometer bracket which is secured by two screws in the bearing block.

POTENTIOMETER ADJUSTMENT

When installing a new potentiometer, it is necessary to orient the potentiometer electrically with respect to the main switch shaft. Care should also be taken to assure that the potentiometer shaft will turn freely after its bracket has been secured to the bearing block. Check Fig. 5 to obtain the proper resistance curve for the switch.

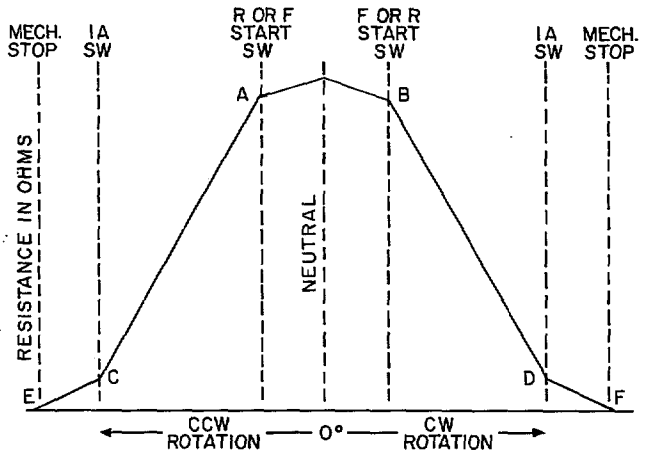


Fig. 5. Resistance curve

To adjust the potentiometer loosen the set-screw on the pinion gear and, with switch in neutral, set the potentiometer for maximum resistance of the potentiometer. Tighten the set screw. Move the shaft slowly from neutral position to full-on position in either direction. Observe the resistance, using a VOM, as switchettes operate at positions A or B, and at positions C or D. If resistance at these points does not agree with table below, reset the potentiometer at neutral and repeat the test. For the hand-operated type switch, it is important to try and balance the resistance in both forward and reverse directions for correct truck performance.

If the potentiometer is suspected of grounding, check as follows: Disconnect all incoming wires and check all terminals to the potentiometer case with a VOM on RX10,000 scale. A reading of infinity (1,000,000 ohms plus) should be obtained.

*Registered trade-mark of the Acheson Colloids Company.

SWITCH	SETTINGS IN OHMS AT DEGREES OF ROTATION			
	A or B	C or D	E or F	
			Foot Operated	Hand Operated
3012BH600-699				
PR Relay Plug	180K-250K @ 7°	Max. 300 @ 27°	Min. 1 @ 36°	Min. 1 @ 30°
Static Plug	3.8K-6.0K @ 7°	Max. 200 @ 27°	Min. 1 @ 36°	Min. 1 @ 30°
3012BH700-729				
Static Plug	3.8K-6.0K @ 7°	Max. 200 @ 28°	Min. 1 @ 36°	Min. 1 @ 30°
3012BH730-769				
Static Plug	3.8K-6.0K @ 7°	Max. 200 @ 28°	Min. 1 @ 36°	Not Available

NOTES:

1. *For foot-operated types, use only the left or right side of the Fig. 5 diagram for CW or CCW rotation.*
2. *For hand-operated types use both the left and the right side of the Fig. 5 diagram.*
3. *Resistance of the potentiometer at positions E or F should never be allowed to read zero ohms, or to pass through zero ohms.*